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When global gating is enough: Admission-time hubness control in anisotropic vector retrieval systems

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ABSTRACT

Vector *hubness*, the tendency of a few points to be nearest neighbours of disproportionately many queries in high-dimensional space, is an attack surface for retrieval-augmented generation: one crafted document can be retrieved for many unrelated queries. The prevailing defence is periodic detection, which leaves an exposure window and requires corpus-wide rescans. We study *admission-time* control, scoring each candidate against sentinel queries and refusing it before it becomes retrievable, and ask whether that control can be maintained incrementally.

Two results follow. First, a single *global* gate suffices. On a 100,000-document BEIR corpus, with attacker anchors disjoint from the defender's sentinels, it recovers effective adversarial hubs at recall 1.0 (AUROC ≈ 1.0) at an $\approx 1\%$ false-positive rate on in-domain general documents, and averages 0.91 ± 0.07 recall on gradient-realised attacks. Second, refining the gate to be *domain-aware* adds nothing: the per-topic marginal over global is $+0.000$ across five encoders, two corpora and two concept definitions. We give a geometric account: anisotropic embeddings couple topic-local and global visibility, so a hub cannot be topic-loud yet globally quiet.

The thresholds are maintained incrementally, exactly and at a per-write cost independent of corpus size. On a real HNSW index admission adds $\approx 3.1\%$ to ingestion latency, its scoring cost is flat to $N = 10^6$, and 1.2% of admit/quarantine decisions flip under approximate indexing, none of them attacks; results reproduce on x86 server hardware. We advocate a preventive global admission gate, with provenance as a complementary control for the tight-domain residual.

1. Introduction

This paper asks a systems-security question with a clean answer: to stop hubness-based poisoning of a retrieval store, is a *single global* admission check enough, or must the defence be specialised per topic? We show that one global check suffices across the settings we evaluate, explain geometrically why, and demonstrate that it is cheap enough to run inline on every ingested document.

Retrieval-augmented generation (RAG) (Lewis et al., 2020; Guu et al., 2020) grounds a language model on documents fetched from a vector store by embedding similarity. The retrieved passages are treated as trusted context, so the store is a security boundary: a document retrieved for a query can steer the model's output for that query. *Hubness*, the concentration of reverse-nearest-neighbour (reverse-kNN) counts on a few points in high-dimensional spaces (Radovanović et al., 2010), turns this into an attack surface. An adversary who crafts a *hub* document can have it retrieved across many unrelated queries, so a

single injected record influences a large fraction of interactions (Zhang et al., 2024; Zou et al., 2025; Zhong et al., 2023).

The prevailing defence is detection: periodically scan the index, estimate reverse-kNN influence, and remove statistical outliers (Habler et al., 2026). Detection after the fact has two structural costs: an *exposure window* between a hub's insertion and the next scan, during which it hijacks queries, and the expense of rescanning a large store. This paper asks whether the same statistic can be applied *at admission* (evaluated on each document *before* it becomes retrievable, so a hub never enters the index) and whether that control can be maintained *incrementally* per write rather than by repeated full rescans. A useful gate must also separate adversarial hubs from *legitimately* general documents (glossaries, overviews) that are naturally hub-like, or it would break normal ingestion.

A natural refinement is to condition the statistic on topic ("domain-aware" detection), on the intuition that a concept-specific hub may be

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loud within one topic yet quiet globally and thus slip past a global threshold. Our central empirical finding is that, on dense single-vector retrieval, this refinement buys nothing: across the evaluated encoders, corpora, attacks, and operating points, domain-aware gating provides no statistically reliable benefit. We explain this geometrically: dense sentence embeddings are *anisotropic*: rather than spreading over the unit sphere they occupy a narrow cone, so cosine similarities share a large common component (Ethayarajh, 2019; Gao et al., 2019), and this coupling means a document that is hub-like within a topic is, with few exceptions, hub-like globally as well, which is why a single global threshold suffices. We show, further, that the global gate is cheap enough to run synchronously on the ingestion path of a real approximate-nearest-neighbour (ANN) index.

Contributions.

1. A *preventive* admission-time formulation of hubness control that evaluates documents before they are retrievable, removing the exposure window left by detective scanning (Sections 3, 5).
2. A *scalable incremental-maintenance* scheme for the admission thresholds that is exact against full recomputation and whose per-write cost is independent of corpus size (amortised for deletion), together with a real-index systems evaluation (HNSW; end-to-end latency, scalability to $N = 10^6$, approximate-index robustness, concurrency) showing the gate adds negligible relative overhead in the evaluated ingestion pipeline (Sections 5, 9).
3. A *reproducible negative result*: a domain-aware per-topic gate provides no measurable benefit over the global gate across five encoders (384–1024 d; MiniLM/BGE/GTE/E5), two compositionally distinct corpora, two concept definitions, and a five-seed sweep (Sections 7, 11).
4. A *geometric explanation* (embedding anisotropy couples local and global visibility) that provides a geometric account of the observed sufficiency of global gating and identifies a tight-domain exception (Sections 8, 10).

2. Threat model and assumptions

System. A vector store holds unit-normalised embeddings of single-passage documents under a fixed encoder; queries retrieve the top- k by cosine similarity (equivalently inner product on normalised vectors). **Adversary.** The attacker injects documents through the ingestion path but cannot alter the encoder, other stored vectors, or the retrieval code. The goal is a hub retrieved for many *anchor* queries the attacker cares about. We use two role-labelled, disjoint query sets throughout, and never interchange them: *sentinels* are the *defender's* probe queries against which the admission gate scores each candidate document, and *anchors* are the *attacker's* target queries a hub is optimised to be retrieved for. We grant the attacker knowledge of the defence and the ability to optimise hubs by gradient methods, and, as the realistic evasion case, hold the attacker's anchor queries *disjoint* from the defender's sentinels (measured overlap 0.000). **Out of scope.** Targeted single-query poisoning (Zou et al., 2025; Ben-Tov and Sharif, 2025), multi-vector/late-interaction retrieval (Khattab and Zaharia, 2020), and model-internal poisoning are out of scope and revisited as limitations.

3. Admission-time global gate

Let the corpus be $C = \{x_1, \dots, x_N\}$, $x_i \in \mathbb{R}^d$, $\|x_i\| = 1$, and let $Q = \{q_1, \dots, q_M\}$ be sentinel-query embeddings. For each sentinel q_j define a threshold τ_j as the similarity of its k th nearest neighbour in C , so a candidate d would enter q_j 's top- k iff $\langle d, q_j \rangle > \tau_j$. The *global hub score* is the fraction of sentinels the candidate would displace into,

$$h(d) = \frac{1}{M} \sum_{j=1}^M \mathbf{1}[\langle d, q_j \rangle > \tau_j], \quad (1)$$

Algorithm 1 Candidate scoring and admission

Require: candidate d , sentinels Q , thresholds $\{\tau_j\}$, budget θ

```

1:  $h \leftarrow \frac{1}{M} \sum_j \mathbf{1}[\langle d, q_j \rangle > \tau_j]$ 
2: if  $h \leq \theta$  then
3:   return ADMIT
4: else
5:   return QUARANTINE
6: end if
```

Algorithm 2 Incremental threshold maintenance (insert / delete)

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1: procedure INSERT( $d$ )
2:   for  $j$  with  $\langle d, q_j \rangle > \min B_j$  do splice  $\langle d, q_j \rangle$  into  $B_j$ , keep top- $B$ ;
    $\tau_j \leftarrow B_j[k]$ 
3: end for
4: end procedure
5: procedure DELETE( $x$ )
6:   for  $j$  such that  $x \in B_j$  (scan,  $O(MB)$  total) do drop  $x$  from  $B_j$ 
7:     if  $|B_j| < k$  then refill  $B_j$  by one rescan ▷ rare
8:     else  $\tau_j \leftarrow B_j[k]$ 
9:     end if
10:  end for
11: end procedure
```

and the gate admits d if $h(d) \leq \theta$, quarantining it otherwise, with θ calibrated to a target false-positive budget on held-out natural documents. A normal single-topic document beats only a small fraction of sentinels; being retrieved by many is the adversarial signal.

4. Domain-aware alternative and geometric hypothesis

The natural refinement conditions on topic. With per-topic local hub rate $s_i(d)$ over topic i 's sentinels, a worst-topic robust- z score

$$z(d) = \max_i \frac{s_i(d) - \text{med}_i}{\max(1.4826 \text{MAD}_i, 1/n_i)}$$

(here MAD_i is the median absolute deviation; the scale floored at the rate resolution $1/n_i$ avoids a degenerate zero-MAD blow-up) is intended to catch a hub that is quiet globally but loud within one topic. The *geometric hypothesis* we test is that, under anisotropy, no such hub exists, so $z(d)$ carries no information beyond $h(d)$ (Section 8).

5. Incremental systems architecture

The gate sits synchronously on the ingestion path: admitted documents are indexed, flagged documents quarantined for review. An external, slowly-updated baseline supplies θ and resists a dilution attack (mass injection that shifts the corpus's own statistics); a velocity monitor watches for many sub-threshold documents converging on one region; a periodic detective scan remains as a backstop for queries outside the sentinel set. Each sentinel maintains its current top- B similarities ($B = k + c$) in a buffer with row ids, so τ_j is the k th entry and scoring is one pass over the sentinels (Algorithms 1–2).

Complexity. Scoring and insertion cost $O(Md)$ and $O(Md + a \log k)$ (a = displaced sentinels); buffered deletion costs $O(MB)$ for the membership check with an occasional $O(N)$ refill that amortises to a constant (a sentinel underflows only every $\sim (B - k)$ buffer hits, and the buffer-hit rate is $\sim 1/N$). Insertions and ordinary deletions are N -independent; deletion is amortised N -independent under the stated random-delete workload, with an $O(N)$ worst-case refill. Per-write costs grow with the sentinel count M and embedding dimension d (and, under an approximate index, with the ANN query used to maintain τ). Space is $O(MB + Md)$. The maintained thresholds are *exact*: on the real bge-large embeddings at the *deployed* sentinel count ($M = 5571$,

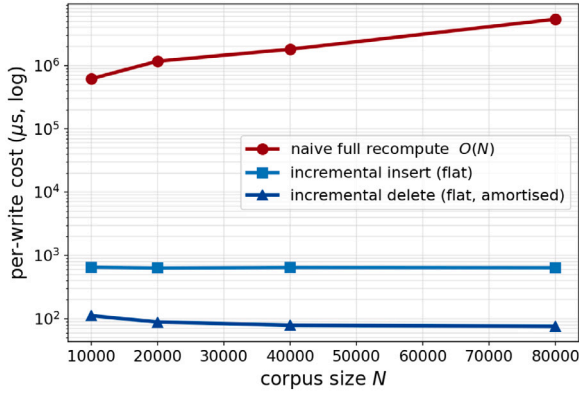


Fig. 1. Incremental threshold maintenance on the real bge-large embeddings: per-write insert/delete cost is flat in N and exact against full recomputation, whereas the naive recompute is $O(N)$.

Table 1

Experimental configuration at a glance.

Component	Setting
Corpus	$N = 100,000$ docs, four BEIR collections (FiQA/TREC-COVID/SciFact/NFCorpus); 10,200 grounded queries
Encoder (headline)	bge-large-en-v1.5 (1024d); L2-normalised (cosine = inner product)
Encoders (sweep)	bge-base, e5-large, gte-large, MiniLM-L6 (384-1024d)
Topics/sentinels	$T = 500$ k -means topics; $M = 5571$ sentinels (eval-query halves + centroids)
Search	Exact k NN ($k = 10$) for separation; real HNSW for the systems study
Calibration	θ frozen at 1% FPR on a <i>disjoint</i> 5000-doc benign set
Attacks	Universal (24); concept (λ -sweep); HotFlip-through-BGE (text-constrained)
Runs	Five seeds {20260612, 7, 42, 123, 2024}; paired bootstrap CIs
Hardware	Apple-silicon (arm64), MPS backend

$k = 10$, $B = 50$; harness parameters in Appendix) the per-write update is flat in N (≈ 0.72 ms insert, ≈ 0.08 – 0.11 ms delete from $N = 10^4$ to 8×10^4) with maximum error 0 and zero buffer refills, while a naive full recompute grows linearly (0.56 s $\rightarrow$ 4.2 s); see Fig. 1. Section 9 evaluates the gate end-to-end on a real approximate index.

6. Experimental methodology

Objectives. The evaluation answers three questions. (Q1, **containment**) does the global gate quarantine adversarial hubs at admission while admitting genuinely-general documents? (Q2, **the negative result**) does conditioning the gate on topic (domain-aware) add anything over a single global threshold? (Q3, **cost**) is the gate cheap enough to run synchronously on a real ingestion path? Q1–Q2 are answered by the separation study (Sections 7–8) and Q3 by the systems study (Section 9). *All headline results are averaged over five seeds with paired bootstrap confidence intervals* (Section 11); Table 1 summarises the configuration. One design principle governs the populations described below: a gate is only interesting if it is tested against benign documents that are themselves genuinely hub-like, so we calibrate and evaluate against organically popular and definitional passages drawn from the corpus itself rather than against random documents alone, and we freeze the operating threshold on a disjoint set before any attack is scored. Implementation particulars are collected in Appendix.

Corpus. $N = 100,000$ documents from four BEIR collections (Thakur et al., 2021), gold-prioritised then subsampled: FiQA (finance;

the concept target; 57,318), TREC-COVID (33,866), SciFact (5183), NF-Corpus (3633); passages under 80 characters dropped; 10,200 grounded queries; 328/500 topics contain a query. **Encoder.** BAAI/bge-large-en-v1.5 (dim 1024) (Xiao et al., 2023), a widely-deployed retrieval encoder of the class these stores run in practice. Embeddings are L2-normalised once, so cosine similarity and inner product coincide throughout; this invariant is what lets the gate, the index, and the attacks all operate in a single geometry. **Topics.** $T = 500$ k -means clusters over the L2-normalised embeddings provide the topic partition that the domain-aware gate conditions on; a query’s topic is the majority topic of its gold documents. **Search.** Exact brute-force k NN ($k = 10$) for the separation study; the systems study (Section 9) uses a real Hierarchical Navigable Small World (HNSW) index (Malkov and Yashunin, 2020). τ_j is each sentinel’s k th-NN similarity over the clean corpus. **Sentinels.** $M = 5571$: held-out evaluation queries (half of each topic’s queries) plus the 500 cluster centroids. **Splits.** Within each topic, queries are split evenly into sentinels and attacker anchors; sentinel-anchor overlap 0.000.

Negatives. The headline natural-negative class is $N\text{-organic} \cup N\text{-curated}$: 1000 *organic* natural hubs (top-1% by reverse- k -occurrence over the queries) and 1500 *curated* in-domain definitional/overview passages mined from the corpus (excluding organic). Two sanity floors: 1500 cross-domain Simple-English-Wikipedia passages and 2000 random documents. A disjoint benign calibration set (5000 random documents) supplies the robust- z baseline; attacks and evaluation negatives are never used for calibration. **Calibration.** The operating threshold θ is selected to give a 1% false-positive rate (FPR) on the *disjoint* 5000-document benign calibration set (never the evaluation negatives), then frozen; we report attack recall and the per-population false-positive rate at that fixed threshold (Table 3). For corpus-derived benign candidates (organic, curated, random, calibration), admission scores use leave-one-out thresholds, so each document is evaluated as a pre-insertion candidate; external candidates (injected attacks, cross-domain Wikipedia) use the standard threshold.

Attacks. Universal hubs (24, random anchor subsamples); concept hubs by a gradient hub-rate search on the unit sphere across a suppression sweep $\lambda \in \{0, 0.5, 1, 2, 4, 8, 16\}$ (Table 4 lists a representative subset; 8 targets, 5 seeds; an effective concept hub retrieves $\geq 20\%$ of its target topic), with a closed-form mean-cosine attack as reference; and HotFlip-through-BGE (Ebrahimi et al., 2018) text-constrained hubs (4 targets $\times$ 5 seeds, length 32, 120 steps), realising the $\lambda^* = 2$ direction as valid but non-fluent token sequences. The Adversarial Hubness Detector (Habler et al., 2026), in its open-source implementation (Cisco AI Defense, 2026), is run as a baseline on the same corpus, sentinels, and hubs.

Reproducibility. Five seeds {20 260 612, 7, 42, 123, 2024} reuse the cached corpus embedding and re-randomise everything downstream that drives the verdict. **Software.** Library versions, encoder model cards, and the remaining configuration particulars are recorded in Appendix.

7. Security effectiveness

We evaluate four security-relevant properties of the gate, each reported over the five-seed protocol of Section 6:

1. **Attack containment (recall).** The fraction of adversarial hubs quarantined at admission, reported separately for universal, concept (λ -suppression sweep), and realised HotFlip-through-BGE hubs (universal/concept recall: the footer of Tables 3 and 4; realised-attack recall: Table 8 and Fig. 8).
2. **Benign preservation (per-population FPR).** The false-positive rate at the frozen 1%-FPR threshold, reported per benign population (organic natural hubs, curated in-domain, cross-domain Wikipedia, and random), so admission of genuinely-general content is separated from the natural-hub residual (Table 3).

Table 2

Benign false-positive floor (no attacks). Scores: global is a rate; per-topic and rknn-z are robust z-scores.

Gate	N-organic p50/p99	N-curated p99	Organic-vs-random AUROC
Global	0.0011/0.0029	0.0007	0.992
Per-topic	3.0/9.0	3.0	0.965
rknn-z	6.0/16.0	4.0	0.992

Table 3

Leakage-free, leave-one-out benign false-positive rate per population: the operating threshold is frozen on the disjoint 5000-document calibration set (global gate, $\theta = 0.0007$); corpus-derived candidates are scored pre-insertion (leave-one-out). Genuinely-general documents are admitted cleanly; organic natural hubs are not separable from adversarial hubs by the statistic.

Benign population	N	FP	FPR
Organic natural hubs (top-1% rev-k)	1000	862	0.862
Curated (definitional/overview)	1500	16	0.011
Random corpus documents	2000	24	0.012
Cross-domain Wikipedia	1500	0	0.000

Attack recall at the same frozen threshold: universal 1.000, concept 1.000.

- Domain-aware marginal (the negative result).** Whether a per-topic gate improves containment or preservation over the global gate, with paired bootstrap confidence intervals (Section 11).
- Decision robustness under approximate indexing.** The fraction of admit/quarantine decisions that change when the threshold is maintained against a real HNSW index rather than exact kNN (Section 9).

Property 4 (and the sentinel-statistic faithfulness) is quantified in the systems study; Properties 1–3 are established here and in Sections 8–11.

Benign floor (no attacks). Curated in-domain passages are statistically indistinguishable from random (curated-vs-random area under the ROC curve, AUROC, 0.518); genuinely-general documents are not hub-like; the cross-domain Wikipedia set is below random (0.414). Organic natural hubs are separable from random (AUROC 0.992): they genuinely behave like hubs. Table 2 reports the three gate variants. With the threshold frozen on the disjoint calibration set (Section 6), the false-positive rate splits sharply by population (Table 3): genuinely-general documents are admitted cleanly (curated 1.1%, random 1.2%, Wikipedia 0.0%), but *organic natural hubs are flagged at 86%*. This is not a tuning artefact: an organic natural hub and an adversarial hub are both hub-like, so the statistic alone cannot separate them. The gate therefore admits genuinely-general content cleanly and detects all universal and concept hubs at the pre-specified decisive operating point (recall 1.0); organic natural hubs fall in the flagged set and constitute the residual that provenance (Section 10) must adjudicate.

Global vs. domain-aware (embedding-space attacks). Across the suppression sweep (Table 4, Fig. 2) per-topic recall equals global recall for every effective hub and never exceeds it; the maximum marginal is +0.000. At $\lambda = 8$ the hub already retrieves only 7% of its target topic, so it has ceased to be effective, and there per-topic (0.342) is below global (0.355). On the embedding-space attacks the global gate reaches recall 1.0 on universal hubs and on concept hubs at the decisive operating point λ^* (AUROC 1.0; ≥ 0.92 across the effective λ range, Table 4).

Fig. 3 shows the decisive score distributions under the global gate: adversarial hubs separate cleanly from genuinely-general documents, while naturally-occurring hubs overlap with the adversarial population and require provenance-based adjudication.

Baselines. On the same corpus, sentinels, and hubs the gate matches the reverse-kNN detector (Habler et al., 2026) at recall 1.0/AUROC 1.0 (Table 5); in our reproduction on this corpus both the global and domain-aware modes are recall-saturated, so this setting has no headroom to exhibit a domain-aware advantage; the detector's own paper reports settings where domain-scoped scanning recovers targeted

Table 4

Gradient concept attack across the off-cluster suppression sweep λ (embedding-space). Concept recall is 1.0 at the pre-specified decisive point $\lambda^* = 2$; the minimum over the effective range ($\lambda \leq 4$, target hit-rate $\geq 20\%$) is 0.925.

λ	Global rate	Target hit-rate	Recall global	Recall per-topic
0	0.211	0.93	1.000	1.000
2 (λ^*)	0.066	0.59	1.000	1.000
4	0.011	0.25	0.925	0.925
8	0.000	0.07	0.355	0.342
16	0.000	0.00	0.000	0.000

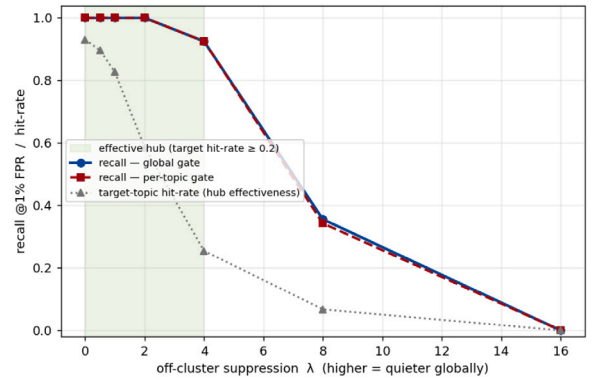


Fig. 2. Global and per-topic recall coincide for every effective hub; per-topic never exceeds global and falls below once the attack is ineffective. Target hit-rate declines with the global hub rate as λ grows, the coupling predicted by anisotropy.

Table 5

Reverse-kNN detector baseline (Habler et al., 2026) on the same data.

Variant	Family	r@0.4%	r@1%	AUROC	Separation
Global	Universal	1.000	1.000	1.000	12.00
Global	Concept	1.000	1.000	1.000	11.56
Concept-aware	Universal	1.000	1.000	1.000	6.22
Concept-aware	Concept	1.000	1.000	1.000	6.22

attacks that evade global detection, consistent with our geometric account that such an advantage appears only where domains are isolated (the tight-domain residual of Section 10). A provenance-only baseline flags every injected hub but also every untrusted natural document, so a provenance-only policy that rejects every untrusted document meets a 1% benign false-positive rate only if at least 99% of benign sources are already classified as trusted.

Realised (HotFlip) attacks. HotFlip-through-BGE hubs reach mean cosine 0.864 (0.84–0.89) of the embedding-space optimum (valid but non-fluent tokens). On the pre-specified run the global gate recovers them at recall 1.0 (AUROC 1.0) and the per-topic gate at 0.93 (AUROC 0.997; this pre-specified seed is not bitwise-reproducible across machines, so we report the five-seed reproduction value of Section 11), at a 1.1% curated-document false-positive rate at the frozen operational threshold. The realised attack is the hardest case for both gates and is analysed across seeds in Section 11.

Robustness to encoder. Table 6 repeats the comparison on five encoders spanning 384–1024 dimensions and four architectures. The per-topic marginal over global is +0.000 on every encoder; the global gate's universal recall is stable throughout (concept recall is 1.0 except on the most anisotropic encoder, e5-large, where it drops to 0.58; that case is hard for *both* gates). On the most isotropic encoder (MiniLM), the per-topic gate's curated-document FPR rises to 0.33 (Table 6) and its calibration is unstable across clusterings while the global gate remains stable, another reason to prefer global. Anisotropy is reported as the mean cosine between per-topic query centroids and the global query centroid; the per-topic marginal does not vary with it (Fig. 4).

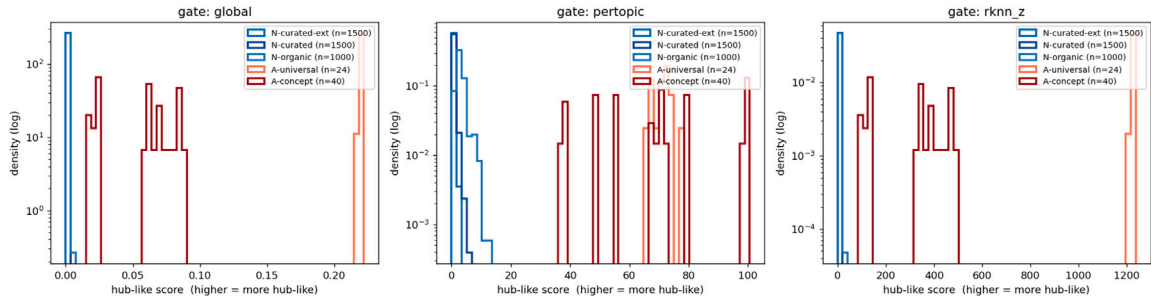


Fig. 3. Score distributions for natural-general negatives vs. universal and concept adversarial hubs. Effective hubs separate cleanly; the natural-general does sit at low scores. This panel is descriptive (score distributions); it is not evaluated at the frozen operating threshold; per-population false-positive rates at that threshold are in Table 3.

Table 6

Per-topic gating adds nothing across encoders (full $N = 10^5$). Anisotropy is the mean topic-centroid-global-centroid cosine; “marginal” is the maximum per-topic-global recall over the λ sweep.

Encoder	dim	Anisotropy	Per-topic marginal	Concept recall (global)	Per-topic curated FPR
all-MiniLM-L6-v2	384	0.466	+0.000	1.000	0.330
bge-base-en-v1.5	768	0.848	+0.000	1.000	0.000
bge-large-en-v1.5	1024	0.860	+0.000	1.000	0.000
gte-large	1024	0.952	+0.000	1.000	0.000
e5-large-v2	1024	0.961	+0.000	0.580	0.000

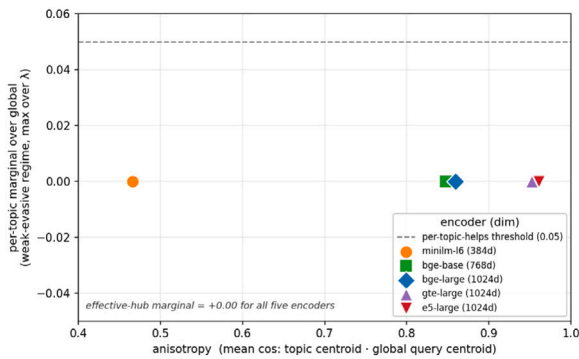


Fig. 4. Across five encoders the per-topic gate’s marginal recall over the global gate is ≈ 0 regardless of anisotropy: domain-aware refinement adds nothing on dense single-vector retrieval.

Robustness to corpus. To test generality beyond the assembled finance/biomedical corpus, we rebuilt the entire pipeline on a compositionally distinct 100,000-document general-web/encyclopedic corpus (NQ, HotpotQA, DBpedia-entity, Quora; 37,033 grounded queries; 8000 sentinels selected as in Section 6, the eval half of each domain’s queries, capped at 7500, plus the 500 cluster centroids). The finding reproduces exactly (Table 7): the benign floor holds (curated-vs-random AUROC 0.514, organic-vs-random 0.996), the global gate recovers every effective concept hub at recall 1.0 (AUROC 0.999) (curated-document FPR 0.9%, organic-hub flag rate 94%, mirroring corpus 1), and the per-topic marginal over global is again +0.000 at every suppression level. On this corpus effective hubs persist to higher λ (the decisive hub at $\lambda^* = 8$ still retrieves 41% of its target topic and is caught at recall 1.0), so the global gate’s sufficiency is, if anything, more pronounced. The negative result is thus robust across two corpora of different domain composition in addition to the five encoders above.

8. Geometric analysis

The negative result is consistent with an anisotropy-based geometric mechanism (we show alignment, not sole causation). In these embedding spaces, per-topic query centroids are positively aligned with the

Table 7

The result reproduces on a second, compositionally distinct corpus (embedding-space attacks, $N = 10^5$, bge-large). Per-topic gating again adds nothing.

Metric	Corpus 1 (finance/biomed)	Corpus 2 (general-web)
Curated-document FPR	0.011	0.009
Random-document FPR	0.012	0.011
Organic-hub flag rate	0.862	0.936
Universal recall	1.000	1.000
Concept recall at λ^*	1.000	1.000
min. recall across effective λ	0.925	1.000
Per-topic marginal over global	+0.000	+0.000

global query centroid (mean topic-centroid-global cosine 0.47–0.96; strong in four of five encoders), so the component of a candidate that raises its topic hub rate also raises its global hub rate: a hub cannot be topic-loud and globally-quiet at once. A direct vector-space optimisation over the unit sphere, maximising topic hub rate subject to a global-rate ceiling, did not find a topic-loud/global-quiet vector in the evaluated searches, and the target-topic hit-rate falls in lock-step with the global rate as λ increases (Table 4). This anisotropy is the well-documented narrow-cone geometry of contextual and sentence embeddings (Ethayarajh, 2019; Gao et al., 2019; Mu et al., 2018; Li et al., 2020; Gao et al., 2021); our result connects it to a security consequence: where the cone is narrow, a single global threshold is sufficient and a domain-aware statistic is redundant.

9. Systems evaluation

Whether admission control is deployable is an architectural question rather than a micro-benchmark one: the gate is viable only if its cost is small beside the work an ingestion path must perform anyway (encoding the document and inserting it into the index) and only if that margin does not erode as the store grows. We therefore evaluate the gate end-to-end on a *real* approximate index (HNSW via hnswlib (Malkov and Yashunin, 2020)) over the real bge-large embeddings, reporting each gate cost as a share of the full pipeline and tracking its behaviour as N scales.

End-to-end ingestion. Per-document ingestion has four synchronous stages: encode 26.4 ms, candidate scoring 0.18 ms, threshold maintenance 0.72 ms (incremental τ update at the deployed $M = 5571$; p95 1.1, p99 1.4 ms), and HNSW insert 2.0 ms (encode/scoring/insert p99 51.6/0.32/3.8 ms), a full ingestion path of 29.3 ms. The *admission-control overhead* (candidate scoring plus synchronous threshold maintenance) is $(0.18 + 0.72)/29.3 \approx 3.1\%$, of which candidate scoring alone is 0.6%. The four stages are timed independently (scoring: the sentinel matmul and threshold compare; maintenance: the buffer update; insertion: the HNSW add) and reported as non-overlapping medians, so the 3.1% does not double-count. Bulk index construction (hnswwlib's multi-threaded `add_items`) runs at 6900 docs/s and adds 440 MB for 100k vectors.

Scalability. We separate two claims. *Exact* incremental threshold maintenance was evaluated through $N = 8 \times 10^4$ (Section 5, Fig. 1). Candidate-scoring and HNSW-insert latency were probed on an index grown to $N = 10^6$: real corpus vectors through $N = 10^5$, then synthetically extended for insert-timing only (Fig. 5, left); the gate's per-write scoring latency shows no trend with index size (139–273 μ s, median 166 μ s), while HNSW insertion grows sub-linearly (1.4 ms $\rightarrow$ 6.2 ms). The gate's N -independence, the design's central claim, holds on a real index, not just in the data-structure model of Section 5. Peak resident memory at $N = 10^6$ is 9 GB.

Generalisation beyond the measurement platform. RAG backends are typically served on x86 clusters with discrete GPUs rather than on the Apple-silicon (arm64, MPS) laptop used above, so we re-ran the full real-HNSW systems and concurrency benchmarks on server-grade x86 hardware: a cloud instance exposing 30 vCPUs of an Intel Xeon Platinum 8358 (a 32-core/64-thread Ice Lake part) with a discrete NVIDIA A10 GPU, using the identical corpus, sentinel set, and 1%-FPR operating point. The eight-thread concurrency results reported below therefore sit well inside the instance's vCPU allocation. Both load-bearing conclusions hold, and the absolute latencies are *faster* than on the laptop, confirming that the laptop figures are a conservative bound.

(i) *N -independence:* as expected from the fixed $M \times d$ sentinel matmul, which carries no N term, per-write candidate-scoring latency shows no trend with index size across $N = 10^4$ to 10^6 , varying over 45–114 μ s (median 49 μ s; the single 114 μ s probe at $N = 2.5 \times 10^5$ is measurement noise rather than an N effect) against 139–273 μ s on MPS, while HNSW insertion grows from 0.66 to 2.8 ms.

(ii) *Negligible relative overhead:* the per-document stages are encode 20.25 ms, candidate scoring 0.093 ms, threshold maintenance (insert path, re-measured on the same instance type) 0.070 ms (flat over 0.069–0.072 ms from $N = 10^4$ to 8×10^4 , exact against full recomputation with zero buffer refills), and HNSW insert 1.16 ms, a full ingestion path of 21.6 ms. Composed exactly as for the laptop, admission control (candidate scoring plus synchronous threshold maintenance) is 0.75% of that path, of which candidate scoring alone is 0.43% (cf. 3.1% and 0.6% on the laptop). Every share quoted for either platform is the ratio of per-stage medians to the summed-median ingestion path, computed before rounding; the printed per-stage latencies are rounded for readability.

Two scoring latencies appear above for each platform, and they measure the same call under different conditions. The sweep times the scoring call in a tight probe loop, where the 22 MB sentinel matrix stays resident in cache; the end-to-end median times the same call on the live ingestion path, between an encode and an index insert that evict it. The end-to-end figure is accordingly the larger of the two on both platforms (0.093 ms against a sweep median of 49 μ s on x86; 0.18 ms against 166 μ s on MPS). The difference reflects cache residency and per-call overhead, not different work.

Concurrent ingestion also improves: gate+insert scales 6.87 $\times$ across eight threads to 4176 docs/s, exceeding the laptop's 5.9 $\times$ /3523 docs/s. Batched gate scoring is the one figure that does not improve, reaching 92,700 docs/s against 142,000 on the laptop; it is not on the critical path, since HNSW insertion, not the gate, bounds ingestion

throughput on both platforms. Realising that scaling on a many-core host requires the gate's linear-algebra thread pool to be pinned so that concurrent ingestion workers do not oversubscribe cores; the configuration detail and the cost of getting it wrong are given in Appendix. Both platform-invariant claims, flat per-write gate cost and negligible relative overhead, are therefore *measured*, not merely predicted, on server-grade x86/GPU hardware.

Approximate-index robustness. The right question is not how far τ drifts but whether the admit/quarantine *decision* changes when τ is maintained against a real HNSW index instead of exact k NN. With HNSW at recall@10 $\approx$ 0.93–0.96, only 1.2% of decisions flip over all candidates, and *no* attack decision flips (universal and concept recall stay 1.0); the flips concentrate in the borderline organic-hub population (58/1000, 5.8%); curated (5/1500), random (8/2000), and Wikipedia (1/1500) populations flip <0.5% and the attack families not at all (0/24 universal, 0/40 concept) (underlying threshold drift 0.0015 mean, p99 0.017). The gate's decisions are robust to the approximate index that production systems actually run.

Concurrency and throughput. The gate is a matmul and vectorises: batched scoring reaches 142,000 docs/s; HNSW insertion remains the dominant throughput bottleneck, though concurrent gate execution introduces measurable contention at higher thread counts. Under concurrent ingestion (Python threads, each gating and inserting into a shared index), throughput scales 5.9 $\times$ across eight threads to 3523 docs/s, a $\approx$ 29% reduction versus insert-only at eight threads (4951 $\rightarrow$ 3523 docs/s), attributable to Python-level contention and shared-index memory traffic (Fig. 6). The concurrency benchmark runs at $N = 5 \times 10^4$, with gate scoring executed per document inside each worker thread and excluding synchronous threshold maintenance; per-thread throughput therefore reflects that composition rather than the sum of the per-document stage medians reported above.

Faithfulness of the sentinel statistic. The gate approximates a full reverse- k NN scan by a sentinel sample; faithfulness is the Spearman correlation between $h(d)$ and the full reverse- k NN count. With the evaluation configuration (sentinels drawn from the eval *half* of queries, $\approx$ 50% coverage) it is 0.77 (Spearman over a 20,000-document sample; a smaller 4000-document sample gives 0.75); this is a coverage artefact, not a mechanism limit. Sweeping sentinel coverage Q (Fig. 7) raises it monotonically to 0.92 at $Q = 8000$ and 0.98 at full query coverage (a pure subsample reaches 1.000, so the τ mechanism itself loses almost nothing). Candidate-scoring latency scales with M and is 0.18 ms at $M = 5571$, so it remains below 0.5 ms at full query coverage. Total admission overhead, including threshold maintenance (which also grows with the sentinel count), should be considered separately.

10. Tight-domain residual and complementary provenance

Defining concepts as real corpus *domains* (rather than k -means clusters) reproduces the +0.000 per-topic marginal. The exception bounds the central claim: in a tight, distinct, under-sampled domain (TREC-COVID, 50 queries) an effective concept hub can be globally quiet and evade *both* gates, and there the concept-aware variant is *worse* than global because natural in-domain documents are themselves concentrated, raising the within-domain floor. This is consistent with the geometric account: an isolated domain is less collinear with the global centroid. For such residual cases, document *provenance* (source trust) is a promising complementary control, orthogonal to the geometric signal; we report a provenance-only baseline and do not claim an integrated gate-plus-provenance policy as an implemented contribution.

When should an architect escalate from the global gate to provenance? The global gate is the default and is sufficient across the broad, well-sampled corpora studied here; the escalation decision is a per-domain screen an operator can run with quantities already computed during calibration and clustering, requiring no attack knowledge. A domain

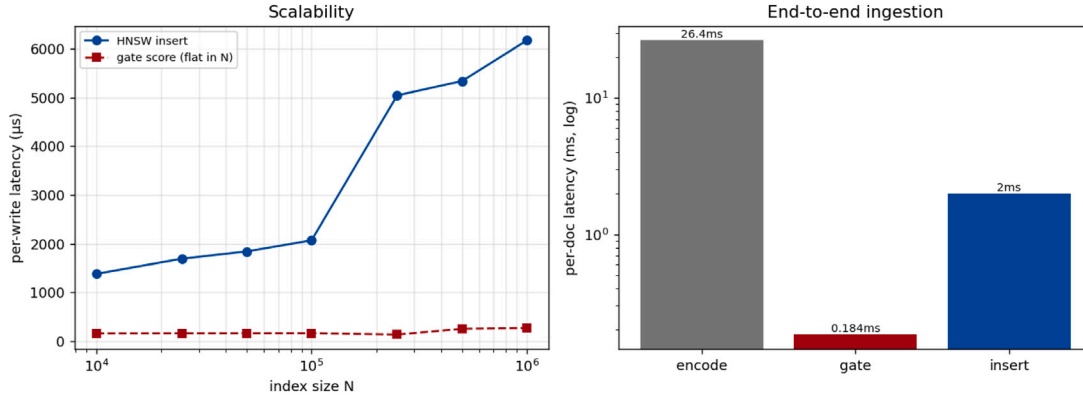


Fig. 5. Real-HNSW systems evaluation. Left: candidate-scoring latency remains flat through $N = 10^6$, while HNSW insertion grows with index size. Right: median ingestion latency for the three stages timed in this benchmark (encode, candidate scoring, HNSW insert); synchronous threshold maintenance is measured separately and reported in the text. Admission control adds $\approx 3.1\%$ to median ingestion latency; candidate scoring alone contributes 0.6% .

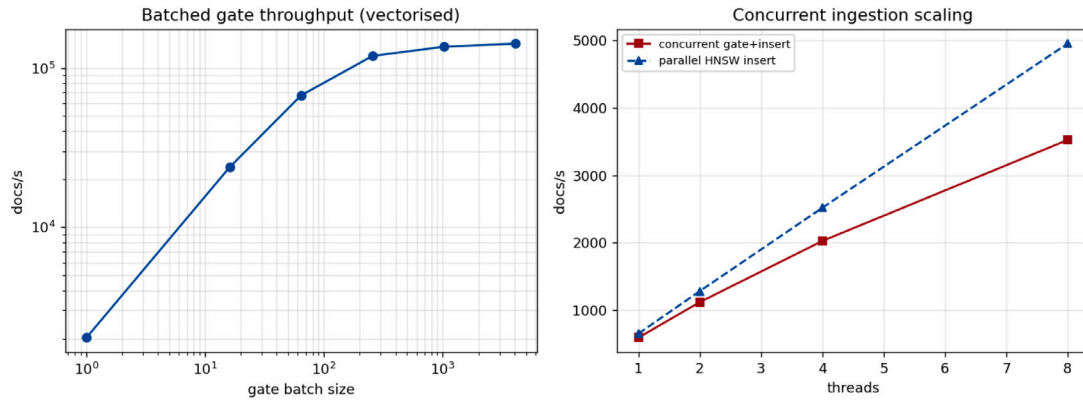


Fig. 6. Left: batched gate throughput (vectorised matmul). Right: concurrent ingestion scales with threads; HNSW insertion remains the dominant operation, although admission control introduces measurable contention at higher concurrency.

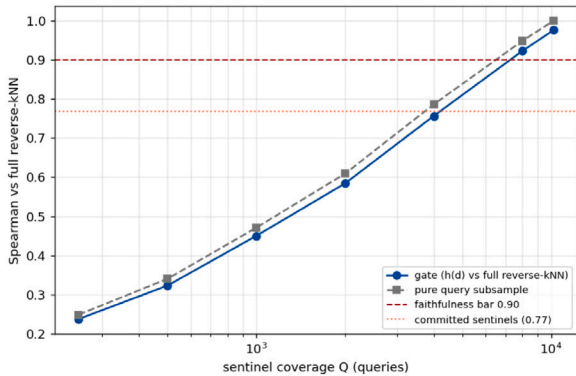


Fig. 7. Gate faithfulness is coverage-limited: Spearman against the full reverse- k NN scan rises with sentinel coverage Q , reaching ≥ 0.90 by $Q \approx 8000$ and 0.98 at full query coverage.

warrants a complementary provenance check only when it is simultaneously (i) *tight*: high within-domain collinearity, i.e. its documents' mean pairwise cosine sits well above the corpus median, so in-domain content already raises the within-domain floor; (ii) *distinct*: its centroid is weakly collinear with the global centroid (low centroid-to-global cosine), the geometric condition under which a concept hub can be locally effective yet globally quiet; and (iii) *under-sampled*: few sentinels cover it (few queries per topic), so the global statistic has little in-domain support. TREC-COVID (50 queries) is our calibrated exemplar

of all three, and is exactly where the concept-aware variant is *worse* than global. Concretely, we recommend flagging domains in, say, the tightest and most distinct deciles of the corpus that also fall below a minimum sentinel-coverage floor; a domain crossing none of the three thresholds is safely handled by the global gate alone.

The escalation is then gated on *consequence*, not geometry alone: even a flagged domain needs provenance adjudication only if its content is trust-sensitive (e.g. clinical, legal, or safety guidance a downstream model would treat as authoritative). *Illustrative scenario.* An operator ingesting a broad web/enterprise corpus runs the single global gate everywhere; a narrow, freshly-stood-up clinical-guidance collection (few curated queries, documents clustered tightly around a handful of protocols, centroid far off the global axis) trips all three screens *and* is trust-sensitive, so ingestion from *that source* is additionally required to carry a trusted provenance signature, while the rest of the pipeline is unchanged. This confines the more expensive, policy-laden control to the small residual where the geometric gate provably loses power, rather than imposing it corpus-wide. Designing and evaluating an integrated gate-plus-provenance policy, including the calibration of the three thresholds on labelled domains, remains future work (Section 13).

11. Reproducibility

The comparison between the global and domain-aware gates rests on two metrics near a decision boundary and sensitive to nondeterminism in clustering, the anchor split, and attack construction. Re-running the downstream pipeline for five seeds (reusing the cached corpus embedding) gives, on the realised HotFlip attack, global recall 0.91 ± 0.07 and per-topic recall 0.85 ± 0.05 (Table 8, Fig. 8). Separately, the

Table 8

Five-seed sweep, realised (HotFlip) attack: per-seed recall and AUROC. The pre-specified seed is the most favourable for the per-topic gate. SDs are population SD (ddof = 0).

Seed	Global recall	Per-topic recall	AUROC _{pt}
20 260 612 (pre-spec.)	1.000	0.930	0.997
7	0.900	0.850	0.975
42	0.900	0.807	0.995
123	0.950	0.850	0.993
2024	0.790	0.800	0.979
mean $\pm$ std	0.908 $\pm$ 0.070	0.847 $\pm$ 0.046	0.988 $\pm$ 0.009

Table 9

Embedding-space suppression-sweep marginal (per-topic – global, max over λ) by seed, a distinct quantity from the realised-attack recalls of Table 8. These values come from the five-seed reproduction sweep, which re-runs clustering, the anchor split and attack construction, so the pre-specified seed's entry here need not equal the committed run of Table 4, which gives +0.000 at every λ .

Seed	emb-space sweep marginal
20 260 612 (pre-spec.)	+0.065
7	+0.000
42	+0.000
123	+0.000
2024	+0.025
mean $\pm$ std	+0.018 $\pm$ 0.025

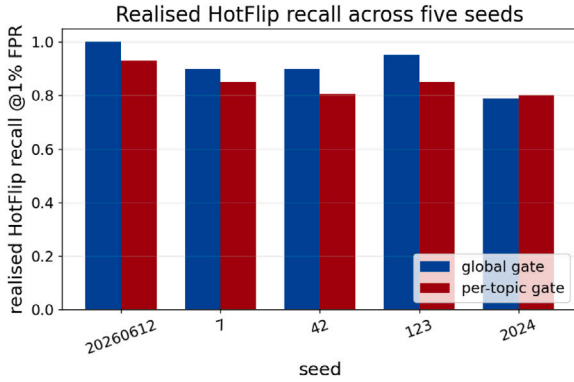


Fig. 8. Five-seed distribution of the realised-attack recall: per-topic never reliably exceeds global, and the pre-specified seed is the most favourable for the per-topic gate.

embedding-space suppression sweep gives a per-topic marginal over global of $+0.018 \pm 0.025$ (Table 9). Neither shows a statistically reliable improvement from domain-aware refinement. The core separation is stable (universal recall 1.000, embedding-space concept recall 1.0 at the pre-specified decisive point (minimum recall 0.925 across the effective suppression range), a 1.1% curated-document false-positive rate at the frozen threshold, attack realisability ≈ 0.86). The pre-specified configuration is the most favourable of the five for the per-topic gate, so we report the verdict conservatively: a single favourable run should not be read as a benefit. A paired analysis (experimental unit: seed, $n = 5$) of the realised-attack (per-topic – global) recall gives a mean of -0.061 (95% bootstrap CI $[-0.091, -0.023]$; Cohen's $d = -1.37$): per-topic is, if anything, *worse*, and the interval excludes zero on the side that favours global. The embedding-space sweep marginal is $+0.018$ (95% bootstrap CI $[0.000, 0.044]$, including zero). A two-sided sign test is underpowered at $n = 5$ ($p = 0.38$); every interval is nonetheless consistent with no benefit from domain-aware refinement. The full pipeline, frozen thresholds, and per-seed artefacts support full replication and are available from the author upon request.

12. Related work

Hubness. The concentration of reverse-nearest-neighbour counts in high-dimensional data was characterised by Radovanović et al. (2010) and mitigated by similarity re-scaling (Schnitzer et al., 2012); we use it as a security signal rather than a retrieval-quality one. **RAG and retrieval poisoning.** RAG (Lewis et al., 2020; Guu et al., 2020) over dense retrievers (Karpukhin et al., 2020; Khattab and Zaharia, 2020; Izacard et al., 2022) is vulnerable to corpus poisoning (Zhong et al., 2023), knowledge-corruption attacks (PoisonedRAG (Zou et al., 2025)), SEO/embedding attacks (GASLITE (Ben-Tov and Sharif, 2025)), and adversarial hubness (Zhang et al., 2024); adversarial text is realised by gradient token attacks (Ebrahimi et al., 2018; Zou et al., 2023). Defences include certified output robustness (RobustRAG (Xiang et al., 2024)), reranking/graph defences (Zheng et al., 2025), and reverse- k NN hubness detection (Habler et al., 2026); provenance and source-trust approaches secure the channel rather than the statistic. We add a preventive, admission-time complement with a negative result on domain-aware refinement. **ANN systems.** Production stores use HNSW (Malkov and Yashunin, 2020), IVF-PQ (Jégou et al., 2011), FAISS (Johnson et al., 2021), ScaNN (Guo et al., 2020), and DiskANN (Subramanya et al., 2019); we show admission is cheap relative to insertion in such an index. **Embedding anisotropy.** Contextual and sentence embeddings occupy a narrow cone (Ethayarajh, 2019; Gao et al., 2019), addressed by all-but-the-top (Mu et al., 2018), whitening/flow (Li et al., 2020; Su et al., 2021), and contrastive training (Gao et al., 2021); encoders used here are BGE (Xiao et al., 2023), E5 (Wang et al., 2022), GTE (Li et al., 2023), MiniLM/SBERT (Reimers and Gurevych, 2019; Wang et al., 2020), benchmarked with BEIR (Thakur et al., 2021)/MTEB (Muennighoff et al., 2023). We tie anisotropy to a security sufficiency condition. **Incremental maintenance and admission control.** Our threshold buffers are an instance of incremental view maintenance (Blakeley et al., 1986; Gupta and Mumick, 1995); admission control is classic in systems (Welsh and Culler, 2003). **Data poisoning.** Training-time and web-scale poisoning (Biggio et al., 2012; Carlini et al., 2024) target learning; we target the retrieval channel at write time.

13. Limitations and discussion

Realised-attack recall. On HotFlip the global gate is 0.91 ± 0.07 , not perfect, and dips to 0.79 on one seed; both gates leave a realised-attack residual. **Tight-domain residual.** An effective hub in an isolated, under-sampled domain can evade both gates (Section 10); a controlled domain-compactness/size sweep and an integrated provenance policy are future work. **Encoder dependence.** Concept-hub separation is encoder-dependent (e5-large at 0.58); the *per-topic-adds-nothing* result is robust across encoders, but absolute separation is not uniform. **Single node.** The systems study measures one machine; multi-node/sharded distributed ingestion is future work. The primary synchronisation challenge is that each threshold τ_j is a *global* k th-NN statistic over the entire corpus, so under sharded ingestion concurrent inserts on different shards must converge on a consistent global threshold before an admit/quarantine decision is final. Enforcing this synchronously reintroduces a serialisation point on the write path (eroding the N -independent per-write cost), whereas relaxing it to eventual consistency reopens a bounded exposure window, recasting the admission-versus-detection trade-off this paper closes on a single node as a distributed-consistency problem, for which approximate or gossip-maintained global thresholds are a natural starting point. End-to-end evaluation against live indirect-prompt-injection payloads is likewise future work. **Attack scope.** Targeted single-query poisoning, multi-vector retrieval, and adaptive low-amplitude/multi-document attacks are out of scope. **Hardware.** Systems measurements were taken on both an Apple-silicon (arm64, MPS) laptop and an x86 cloud instance (30 vCPUs of an Intel Xeon Platinum 8358, with a discrete

NVIDIA A10 GPU); the gate's N -independence and small relative overhead hold on both platforms, with lower per-write latencies and better concurrent scaling on the server, so the qualitative conclusions are not an artefact of the measurement hardware. **Statistical power.** The paired bootstrap and effect size reported in Section 11 rest on $n = 5$ seeds; more seeds, particularly on the realised (HotFlip) attack, and more encoders would tighten the interval, though the point estimates already favour the global gate.

14. Conclusion

A preventive, admission-time *global* statistical gate separates effective adversarial hubs from genuinely-general documents at a low measured false-positive rate (curated 1.1%, random 1.2%, Wikipedia 0.0%), in a realistic evasion setting and on par with a recent detector; naturally-occurring hubs remain statistically ambiguous and require provenance-based adjudication. The gate is maintainable incrementally at a per-write cost that is exact and N -independent (amortised for deletion), and, on a real HNSW index, adds $\approx 3.1\%$ to the ingestion path (candidate scoring 0.6%), with 1.2% of admit/quarantine decisions flipping under approximate indexing (none of them attacks). A domain-aware refinement provides no benefit in any evaluated setting, a result consistent with an anisotropy-based mechanism in which topic-local and global visibility are coupled. We advocate a global admission gate with provenance as a complementary control for the tight-domain residual, and identify multi-node scalability and adaptive-adversary evaluation as the primary next steps.

CRedit authorship contribution statement

Prashant Kumar Pathak: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Tarun Kumar Sharma:** Writing – review & editing, Validation, Resources, Data curation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

LLMs were used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality. The construction, experiments, and reported numbers are the authors' own work; the LLM did not run experiments or generate findings. No proprietary data was shared with the LLM, and the authors take responsibility for the contents of this paper.

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Appendix. Implementation and operational detail

This appendix collects the configuration and implementation particulars required to reproduce the experiments. They are separated from the main text so that the architecture and the experimental design remain in view there; no reported result depends on a choice recorded here beyond its reproducibility.

Software environment. Python 3.11 with torch 2.8, sentence-transformers 5.2 (Reimers and Gurevych, 2019), scikit-learn 1.7, hnswlib 0.8, and numpy/scipy, on arm64 macOS with the MPS backend; the cross-platform validation of Section 9 additionally uses an

x86 cloud instance (30 vCPUs of an Intel Xeon Platinum 8358) with an NVIDIA A10 GPU, running numpy 2.2.6 against OpenBLAS 0.3.29. Encoder model cards: BAAI/bge-large-en-v1.5, BAAI/bge-base-en-v1.5, intfloat/e5-large-v2, thenlper/gte-large, sentence-transformers/all-MiniLM-L6-v2.

Encoding and clustering settings. Embeddings are computed in float32 with max_seq_length 256 and L2-normalised once. The query-side instruction that the bge family expects is prepended to queries only, never to documents. The topic partition uses Mini-BatchKMeans over the normalised embeddings.

Maintenance-benchmark harness. The incremental-maintenance measurement of Section 5 times 200 randomly generated update operations at each tested corpus size, at $k = 10$, buffer size $B = 50$, and the deployed sentinel count $M = 5571$. Insert operations use random unit vectors, so most candidates are not hubs, matching a realistic admission stream; delete operations remove randomly chosen live documents.

Thread-pool pinning on many-core hosts. The gate's per-document cost is a single $M \times d$ matrix product, which the underlying BLAS library will by default parallelise across all available cores. Under thread-level ingestion concurrency this nests a parallel matmul inside each of several ingestion workers, oversubscribing the machine: on the 30-vCPU x86 instance, leaving the BLAS pool unpinned collapses the scaling to $\approx 1.9\times$. Pinning the gate matmul to one BLAS thread per ingestion worker restores the near-linear figure reported in Section 9. The effect is a property of nested thread pools rather than of the gate, but it is the one configuration detail an operator must get right to reproduce the throughput numbers.

Data availability

Data will be made available on request.

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